

Week 7 – Prompt Engineering for Business Intelligence

Intern Name: Abhay Tayal

Intern ID: 26100250 (AI-Enabled Business Intelligence Internship)

Section 1 –BI Prompts

This section contains ten structured Business Intelligence prompts created for different analytical tasks. Each prompt was designed to evaluate how AI can assist in solving common BI problems such as data cleaning, KPI definition, SQL query generation, dashboard summarization, and business recommendations. The AI-generated responses were reviewed and evaluated based on their relevance, clarity, and usefulness for business decision-making.

#	BI Task	Prompt	AI Tools Used	Output Summary	Evaluation
1	Data Cleaning Steps	I have an inventory dataset containing missing values, duplicate records, inconsistent date formats, and incorrect data types. Suggest a step-by-step data cleaning process before performing Business Intelligence analysis.	ChatGPT	Suggested handling missing values, removing duplicate records, standardizing date formats, correcting data types, and validating the cleaned dataset before analysis.	The response was practical and covered the essential preprocessing steps required for preparing a reliable BI dataset.

2	Define Business KPIs	Suggest three important Key Performance Indicators (KPIs) for a food processing inventory management system that focuses on reducing raw material wastage.	ChatGPT	Recommended Inventory Utilization Rate, Inventory Waste Percentage, and Expiring Soon Batches as the primary KPIs.	The suggested KPIs directly supported the business objective and were suitable for monitoring inventory performance.
3	Executive Summary	Write a professional executive summary for a Business Intelligence dashboard that monitors inventory quantity, expiry dates, and production planning.	ChatGPT	Generated a concise executive summary highlighting key metrics, operational insights, and business recommendations.	The summary was well-structured and suitable for management-level reporting with only minor edits required.
4	SQL Query Generation	Generate an SQL query to identify raw material batches that will expire within the next seven days.	ChatGPT	Generated an SQL query using date filtering to retrieve batches nearing their expiry date.	The generated query was logically correct and useful after adjusting it to match the database schema.
5	Chart Recommendation	Recommend the most suitable chart to visualize monthly inventory batch entries and explain why it is appropriate.	ChatGPT	Recommended a line chart to visualize monthly trends and explained its suitability for time-series analysis.	The recommendation matched standard Business Intelligence visualization practices.

6	Correlation Interpretation	Interpret the relationship between inventory quantity and remaining inventory quantity in a food processing inventory dataset.	ChatGPT	Explained the possible positive relationship and highlighted factors affecting inventory consumption.	The interpretation was useful but required validation using actual business data before drawing conclusions.
7	Data Quality Check	Identify possible data quality issues that may affect inventory analysis in a Business Intelligence project.	ChatGPT	Identified missing values, duplicate records, inconsistent formats, invalid dates, and incorrect data types.	The response covered the major data quality issues commonly encountered during BI projects.
8	Slide Headline Generation	Write a professional presentation headline for a dashboard showing inventory expiry analysis and inventory utilization trends.	ChatGPT	Generated a concise business-oriented presentation title suitable for executive reporting.	The headline was professional, clear, and appropriate for management presentations.
9	Business Recommendation	Based on high inventory waste caused by raw material expiry, provide three business recommendations to improve inventory management.	ChatGPT	Suggested implementing expiry alerts, improving inventory monitoring, and optimizing procurement planning.	The recommendations were practical, business-oriented, and aligned with the project objectives.
10	Explain Technical Concept	Explain the concept of Key Performance Indicators (KPIs) to a non-technical business manager in simple language.	ChatGPT	Explained KPIs using simple business examples without technical jargon.	The explanation was easy to understand and suitable for non-technical stakeholders.

--	--	--	--	--	--

Section 2 – Prompt Variation Comparison

This section compares three different prompt-writing approaches for the same Business Intelligence task to evaluate how prompt quality influences AI-generated responses. The objective is to understand the importance of providing context, defining the target audience, assigning an AI role, and specifying output constraints to improve the relevance, accuracy, and usefulness of AI-generated business insights.

Variation 1

Prompt

Write an executive summary for an inventory dashboard.

AI Output

The AI generated a short and generic executive summary highlighting inventory monitoring and overall business performance. However, it lacked project-specific details, measurable insights, and actionable recommendations.

Evaluation

The response was readable but too generic for business reporting. It required significant editing to make it relevant to the HimShakti inventory dashboard.

Variation 2

Prompt

You are a Business Intelligence Analyst. The dashboard shows inventory quantity, remaining stock, expiring batches, monthly batch entries, and average shelf life for HimShakti's food processing inventory system. Write a 150-word executive summary for warehouse managers.

AI Output

The AI produced a more focused executive summary discussing inventory trends, stock monitoring, expiry management, and production planning. The response aligned well with the dashboard context but still included a few generalized statements.

Evaluation

The additional business context significantly improved the quality and relevance of the response. Only minor factual edits were required.

Variation 3

Prompt

Act as a Senior Business Intelligence Consultant. Write a professional executive summary in three sections: Key Findings, Business Impact, and Recommendations. Use business-friendly language, avoid technical jargon, keep the summary within 150 words, and focus only on verified insights from the inventory dashboard.

AI Output

The AI generated a structured executive summary with clear business findings, operational impact, and practical recommendations. The response followed the requested format and was suitable for management-level reporting.

Evaluation

This prompt produced the highest-quality response because it clearly defined the role, audience, structure, and output constraints. The generated content required very few edits before final use.

Comparison of Prompt Variations :-

Among the three prompt variations, the third prompt produced the most accurate and business-oriented response. The first prompt generated a generic summary because it lacked sufficient context. The second prompt improved the quality by providing project-specific information and defining the target audience. The third prompt performed the best because it included a clear professional role, structured output format, business-friendly language, and specific constraints. This comparison demonstrates that well-designed prompts significantly improve the relevance, clarity, and usefulness of AI-generated outputs for Business Intelligence tasks.